Experiment 1: Text Embedding and Semantic Similarity Search

Aim

To generate text embeddings using a pre-trained Sentence Transformer model and perform semantic similarity search to identify sentences that are most similar to a given query.

Algorithm

1. Start the program.
2. Import the required libraries.
3. Load a pre-trained Sentence Transformer model.
4. Define a set of sample sentences.
5. Convert the sentences into numerical embeddings.
6. Enter a query sentence.
7. Generate the query embedding.
8. Calculate cosine similarity between the query and all sentences.
9. Display the most similar sentences and their similarity scores.
10. Stop the program.

Code:

```
!pip -q install sentence-transformers

from sentence_transformers import SentenceTransformer

from sklearn.metrics.pairwise import cosine_similarity

# Load pre-trained model

model = SentenceTransformer('all-MiniLM-L6-v2')

# Sample sentences

sentences = [

    "Artificial Intelligence makes machines intelligent.",

    "Machine Learning helps computers learn from data.",

    "Python is a popular programming language.",

    "Deep Learning uses neural networks.",
```

```
    "Generative AI creates new text and images."
]
# Generate text embeddings
embeddings = model.encode(sentences)
# Input query
query = input("Enter your query: ")

# Generate query embedding
query_embedding = model.encode([query])

# Calculate cosine similarity
scores = cosine_similarity(query_embedding, embeddings)[0]

# Sort results by similarity
results = scores.argsort()[::-1]

# Display top 3 results
print("\nSemantic Similarity Search Results:")

for i in results[:3]:
    print(f"\nSentence: {sentences[i]}")
    print(f"Similarity Score: {scores[i]:.4f}")
```

Input:

Enter your query: How do computers learn from data?

Output:

```
Enter your query: How do computers learn from data?

Semantic Similarity Search Results:

Sentence: Machine Learning helps computers learn from data.
Similarity Score: 0.6936

Sentence: Artificial Intelligence makes machines intelligent.
Similarity Score: 0.4518

Sentence: Deep Learning uses neural networks.
Similarity Score: 0.4467
```

Result:

Thus, text embeddings were successfully generated using a pre-trained Sentence Transformer model, and semantic similarity search was performed using cosine similarity to identify the sentences most relevant to the given query.